

Volume 0 — Exact Witnesses at the Physics Boundary

The former T4–T6 "Projection Craft" papers, rebuilt as standalone R Theory.

What this volume is. Three books that take the canonical half-angle calculus of [Book 0](#) to the boundary of accepted physics — relativistic kinematics, the Dirac spinor, weak helicity, Coulomb structure, color, bound states, and canonical spin–geometry — and report exactly what is found there: exact coordinate correspondences, exact conditional theorems, honest obstructions, and no new physical law.

Why it is standalone. The source material was written as "Projection Craft" essay-books T4, T5, T6, framed as continuations of the T-paper series. That framing is not carried over. Each book here states its own dependencies, its own imports, and its own status ledger from scratch. Nothing is cited to a T-paper authority; every result is presented with the mathematics or the import that supports it.

Dependencies. All three books inherit the Book 0 core: the reciprocal primitive calculus, the one-generator reduction, the Clifford/bilinear structure, and the coframe geometry. Books 21 and 22 additionally depend on the internal paper *Reciprocal Half-Angle Coordinates in the Radial Dirac–Coulomb Problem* (read and algebra-checked 2026-09-19), which supplies the conditional spinor-ratio identities both books use. Named physics imports — special relativity, the Dirac equation, V–A, QCD, Coulomb/Dirac–Coulomb theory, Maxwell, Einstein–Hilbert, Einstein–Cartan — are declared per section, never smuggled.

Provenance note (2026-09-19). The T6 source preamble credits the T5 source with a "Yang–Mills source-free-vacuum obstruction" result — a static homogeneous constant noncommuting ansatz whose curvature the source-free Yang–Mills equations force to vanish. A full-text search of the T5 source (v2 reconstruction) finds no such result: the only occurrence of "Yang–Mills" is the QCD import line. The attribution describes the original T5 research draft, not the reconstructed text, and it is **not repeated** anywhere in this volume. If the original computation is located and verified, it will be added with its source cited; until then the claim does not exist in R Theory.

The shared discipline

Every book in this volume closes with an operational prediction-difference audit. For a fully specified Projection-augmented model MP against its accepted reference $M0$, on a common measurement contract,

$$\Delta_{op}(MP, M0) = \{ c \in C_{exp} : P_P(\cdot | c) \neq P_0(\cdot | c) \}$$

is the set of experimental configurations where the predictions differ. Three terminal statuses are distinguished: **Class A** — fully specified and $\Delta_{op} = \emptyset$ (operationally equivalent); **Class B** — incomplete, Δ_{op} undefined (an open ansatz is not a nonempty prediction set); **Class C** — fully specified with $\Delta_{op} \neq \emptyset$. An invertible reparameterization can never leave Class A; that is proved, not assumed.

The null hypothesis retained by all three books is **C0**: *Projection Craft changes representation but not physics*. It is falsifiable — a future fully specified Class-C model would defeat it — but undefeated by everything in this volume. Exactness without promotion is the theme: the correspondences are real mathematics, and they are not new laws of nature.

The books

Relativistic and Spinor Witnesses

The kinematic witness ($\sin \chi = \beta$, with $sxp/cxp/crx$ as the Dirac half-angle ratio and rapidity weights), the two-state spinor completion and its independent phase, the V–A helicity weights after explicit chiral import, the beta-decay/Coulomb two-angle bridge, and the adversarial electromagnetic comparison — Jones, duality, Newman–Penrose, Veronese, conformal — ending in the full Δ_{op} audit. Every fully specified construction: Class A. $\Delta_{op} = \emptyset$.

[Read Book 20](#) →

Fermion Composition, Color Closure, and Bound-State Geometry

The composition firewall, the three-state carrier rebuilt in the correct logical order ($C^3 \rightarrow A_2 \rightarrow \text{su}(3)$, tetrahedral metric demoted to compatible realization), the QCD import with its spatial firewall, the exact two-fermion invariant, the below-threshold coordinate and its conditional Dirac–Coulomb convergence, observational discipline with verified hydrogen/deuteron/muonium numbers, the frozen failure record (81/32, fifth lattice, musical commas), and a firewalled E8 appendix. $\Delta_{\text{op}} = \emptyset$.

[Read Book 21](#) →

Canonical Spin–Geometry and the Primitive Symplectic Atlas

The volume's strongest exact result: after the declared radial two-component chart and the imported radial Dirac dynamics, the four primitives are canonical charts on one radial-spinor phase plane — $p_{\text{sxp}} = 2F^2$, $p_{\text{srx}} = -2G^2$, $p_{\text{cxp}} = (F-G)^2$, $p_{\text{crx}} = -(F+G)^2$, common form $\omega_{\text{spin}} = 4 \, dF \wedge dG$. Around that core: the Prüfer canonical pair, the first-order spherical defect action, charge and angular momentum as canonical momenta, Kerr/Kerr–Newman half-angle identities with their guards, and the full does-not-establish ledger.

$\Delta_{\text{op}} = \emptyset$.

[Read Book 22](#) →

Relativistic and Spinor Witnesses

Volume 0. Exact conditional correspondences between the half-angle calculus and accepted relativistic, weak, Coulomb, and electromagnetic structures — and the audit that keeps them honest.

About this book. Rebuilt as standalone R Theory from the former T4 essay-book. It asks one question of each imported physical structure: does the canonical half-angle calculus occur inside it exactly, and if so, does anything new follow? The answer, section by section, is *yes, exactly* — *and no, nothing new*. That second clause is not a disappointment; it is the result. Every import is declared, every obstruction is kept, and the book closes with a formal prediction-difference audit.

Status key. **EXACT** proved mathematics · **CONDITIONAL** exact after a declared import or contract · **IMPORT** accepted physics taken as input, not derived · **OBSTRUCTION** a proved or retained negative result.

Depends on: [Book 0](#) (primitive calculus, one-generator reduction); the internal paper *Reciprocal Half-Angle Coordinates in the Radial Dirac–Coulomb Problem*.

§1 Kinematics

§2 Spinor

§3 V–A

§4 Beta/Coulomb

§5 Electromagnetism

§6 Audit

§1 — The relativistic kinematic witness

Take a massive particle moving at speed β . There is a classical angle hiding in its kinematics: the angle χ with $\sin \chi = \beta$ and $\cos \chi = mc^2/E$. On this chart the half-angle coordinate $q = \text{sxp}(\chi) = \tan(\chi/2)$ is simultaneously three familiar things: the free-Dirac lower/upper spinor component ratio, the rational generator of β and γ , and the square root of the light-cone ratio. The reciprocal pair $\text{cxp} = e^{\eta}$, $\text{crx} = e^{-\eta}$ is the rapidity itself — and $\text{cxp} \cdot \text{crx} = 1$ is exactly the normalized mass shell, already imposed by the reference theory.

Kinematic dictionary **CONDITIONAL**

Under the contract $\sin \chi = \beta$, $\cos \chi = mc^2/E$ on the positive-energy branch:

$$\text{sxp}(\chi) = pc/(E+mc^2) = \tan(\chi/2), \quad \text{srx}(\chi) = (E+mc^2)/pc$$

$$\text{cxp}(\chi) = (E+pc)/mc^2 = e^\eta, \quad \text{crx}(\chi) = (E-pc)/mc^2 = e^{-\eta}$$

The mapping is invertible on the declared branch; the reciprocal identity is the mass shell. This is a coordinate theorem after importing special-relativistic kinematics and the free Dirac spinor —

IMPORT both.

OBSTRUCTION **Rank is not created by notation.** Evaluating many functions on the single coordinate χ does not manufacture additional independent physical degrees of freedom. The dictionary above is one angle's worth of content, however many names it carries.

§2 — Spinor completion and the native ellipse

One real coordinate is not a spinor. Lifting $\mathbf{q} = \text{sxp}(\chi)$ to a complete two-state spinor requires an **independent relative phase** ϕ — the one-generator obstruction survives physical completion. The completed state has an exact Bloch vector $(\cos \chi \cos 2\phi, \cos \chi \sin 2\phi, \sin \chi)$, and the reciprocal pair becomes exact component-amplitude odds. The native complex carrier $\Psi = H + iV$ is characterized exactly — an anisotropic real-linear squeeze of the circular phase with the counter-rotating decomposition $\Psi = (i/8)(e^{2ix} + 3e^{-2ix})$ — but without an independently selected physical embedding of its quadratures it determines no unique observable.

Spinor completion **CONDITIONAL**

With independent phase ϕ , the completed spinor is an exact coordinate representation of imported two-state geometry. No new Hamiltonian, interaction, phase law, or observable is supplied. The bare native carrier as unspecified physical embedding is **Class B** — its prediction set is undefined until a complete contract and dynamics are supplied.

§3 — Weak V–A helicity weights and the chirality-orientation

obstruction

Now import the Standard Model's left-chiral charged-current projector $P_L = (1-\gamma^5)/2$ — imported, not derived. For a massive electron the favored/wrong helicity probabilities are $(1\pm\beta)/2$, and their amplitude odds are exactly the reciprocal pair:

$$\sqrt{P_{\text{fav}}/P_{\text{wrong}}} = \text{cxp}, \quad \sqrt{P_{\text{wrong}}/P_{\text{fav}}} = \text{crx}$$

The equality is exact under the §1 kinematic contract. But the calculus is left–right symmetric: the mirror right-chiral orientation is equally available to the internal reciprocal structure until the Standard Model orientation is imported. Projection reconstructs the helicity-weight geometry of an imported chiral interaction; it does not derive why the interaction is left-chiral. That is the **chirality-orientation obstruction**, and it is retained, not repaired.

CONDITIONAL **V–A helicity correspondence.** After importing the left-chiral orientation, cxp/crx are the exact favored/wrong helicity amplitude odds. **OBSTRUCTION** The orientation itself is not derived; a Projection-only derivation of chirality selection is Class B (unspecified).

§4 — Beta decay, Coulomb dressing, and the two-angle bridge

Follow the emitted beta electron into the daughter nucleus's Coulomb field. Two independent physical angles are compulsory — not one. The continuum kinematic angle χ ($\sin \chi = \beta$, from §1) encodes the electron's mass-shell kinematics; the Coulomb structural angle a ($\sin a = Z\alpha$, $\cos a = \sqrt{1-(Z\alpha)^2}$) encodes the daughter's Coulomb coupling. Identifying them would force $\beta = Z\alpha$ and collapse the continuous beta spectrum to a charge-selected velocity — the **two-angle necessity theorem**. The standard Coulomb parameter becomes their exact quotient,

$$\xi = Z\alpha/\beta = \sin a / \sin \chi = u(1+q^2) / [q(1+u^2)]$$

with $u = \text{sxp}(a)$, $q = \text{sxp}(\chi)$. The baseline allowed phase-space measure becomes fully rational in q and the endpoint coordinate q_0 . But the relativistic Fermi function keeps its non-elementary core $\exp(\pi\xi) |\Gamma(\lambda+i\xi)|^2$: the half-angle substitution rationalizes the kinematic and projective layers but

does not generate the complex-Gamma Coulomb scattering dynamics. The structural angle a is the $K=1$ member of the Coulomb circle of the radial Dirac–Coulomb paper — the bound and continuum problems share the chart but use it differently ($E_{\text{bound}}/m = \cos a < 1$ versus $E_{\text{cont}}/m = \sec \chi > 1$).

Two-angle Coulomb structure CONDITIONAL

- a and χ are generically independent; $a = \chi$ would incorrectly force $\beta = Z\alpha$.
- $\xi = \sin a / \sin \chi$ exactly; phase space rational in (q, q_0) .
- The Fermi function, nuclear matrix elements, radiative and finite-size corrections, endpoint energy, and the nuclear scale R remain IMPORT s.
- OBSTRUCTION The complex-Gamma factor lies outside the finite rational primitive closure — the special-function layer is imported dynamics, not derived.

§5 — Electromagnetic comparison: the adversarial audit

This section is deliberately adversarial. The question is not whether half-angle, Cayley, reciprocal, or doubled-angle forms occur in electromagnetism — they do, abundantly. The question is whether any occurrence survives the redundancies of electromagnetic description. The audit walks through nine witnesses and strips each one:

- **Jones polarization:** $z = e^{i\varphi} \exp(\alpha)$ exactly — but phase φ and intensity scale A^2 are independent imported data (coherency/Stokes).
- **Duality:** $2V = \cos 2x$, $4H = \sin 2x$ give an exact $SO(2)$ duality-rotation matrix — but constant duality is an accepted symmetry, not a new prediction.
- **Native carrier:** $\Psi = H + iV$ is an anisotropic squeeze with exact 1:3 counter-rotating decomposition — but direct $H \leftrightarrow E$, $V \leftrightarrow B$ assignment fails an ordinary traveling wave; the **potential-first** contract $A \propto H$ recovers the standard wave after differentiation (a repair that reproduces Maxwell, not a new law).
- **Local duality:** making the duality angle spacetime-dependent gives an $SO(2)$ connection with zero curvature — pure gauge. The scalar alone does not defeat C_0 by being made local.

- **Newman–Penrose:** cpx/crx match reciprocal boost weights exactly — but those are tetrad-gauge weights, not new observables.
- **Null Maxwell spinor:** the degree-two Veronese map $[1:w:w^2]$ — but only the null rank-one sector, not the generic field.
- **Two CP^1 factors:** polarization CP^1 and propagation-direction CP^1 are independent state spaces; one real scalar cannot supply both.
- **Conformal dressing:** in four dimensions the vacuum Maxwell equations are conformally invariant on 2-forms — a scalar conformal factor changes nothing in source-free propagation.
- **Helicity:** electromagnetic helicity (massless spin-1) is not weak chirality (massive Dirac); the $q=1/2$ numerical coincidence (odds 3:1 in both sectors) promotes no cross-sector law.

The repeated verdict: exact coordinate correspondences, no surviving Projection-specific invariant.

OBSTRUCTION **Invariant-survival audit.** After removing projective, gauge, tetrad, duality, and conformal redundancies, every electromagnetic witness above is an exact correspondence inside known representation freedom. None establishes a Projection-specific invariant. C0 is strengthened, not weakened, by the redundancy analysis.

§6 — The prediction-difference audit and book closure

Correspondence is not promotion. For a fully specified Projection-augmented model MP against its accepted reference M0, define the operational prediction-difference set

$$\Delta_{\text{op}}(\text{MP}, \text{M0}) = \{ c \in C_{\text{exp}} : P_P(\cdot | c) \neq P_{\emptyset}(\cdot | c) \}$$

— the experimental configurations where the predictions actually differ. Three terminal statuses: **Class A** (fully specified, $\Delta_{\text{op}} = \emptyset$), **Class B** (incomplete — Δ_{op} *undefined*, and an open ansatz is not a nonempty prediction set), **Class C** (fully specified, $\Delta_{\text{op}} \neq \emptyset$). Two firewall theorems: an invertible representation change can never create a nonempty Δ_{op} , and incompleteness is not novelty.

The audit walks every construction of §§1–5. Each fully specified one — the kinematic dictionary, the completed spinor, the V–A odds, the two-angle beta rewrite, every electromagnetic witness — lands in **Class A**. Each potentially nonredundant but unfinished proposal — bare native-carrier embedding,

Projection-only chirality selection, scalar-only local duality dynamics — lands in **Class B**. **No Class-C candidate is present.**

Candidate status matrix (condensed)

Relativistic primitive dictionary	Class A, $\Delta=\emptyset$
Free-Dirac half-angle witness	Class A, $\Delta=\emptyset$
Completed two-state spinor	Class A, $\Delta=\emptyset$
Native ellipse (bare mathematics)	exact internal theorem
Native ellipse (physical embedding) ..	Class B, Δ undefined
V–A helicity-weight correspondence ...	Class A, $\Delta=\emptyset$
Projection-only chirality selection ..	Class B, Δ undefined
Beta/Coulomb two-angle rewrite	Class A, $\Delta=\emptyset$
Jones/coherency witness	Class A, $\Delta=\emptyset$
Constant Maxwell duality	Class A, $\Delta=\emptyset$
Potential-first Maxwell wave	Class A, $\Delta=\emptyset$
Scalar-only local duality	pure-gauge Class A / failed symmetry
NP reciprocal boost weights	Class A, $\Delta=\emptyset$
Null Maxwell spinor / Veronese	Class A, $\Delta=\emptyset$
Conformal scalar dressing	Class A, $\Delta=\emptyset$
Class C candidates	none

Book 20 closure **EXACT**

The canonical calculus repeatedly furnishes exact, unusually compact adapted coordinates on standard relativistic, spinorial, weak-helicity, Coulomb, and electromagnetic structures. By explicit obstruction and operational audit, the presently specified correspondences do not alter the invariant predictions of their accepted reference theories. Accordingly **C0 — Projection Craft changes representation but not physics** — is retained as the undefeated null hypothesis, and $\Delta_{\text{op}} = \emptyset$ for every fully specified construction in this book. This is a theorem about the content of Book 20, not a universal no-go: C0 remains falsifiable by a future fully specified Class-C model with a stated measurement contract.

Strongest positive results retained: the one-contract kinematic dictionary ($\text{sxp} \leftrightarrow \text{Dirac ratio}$, $\text{cxp} \leftrightarrow e^\wedge \eta$, $\text{cxp} \cdot \text{crx} = 1$ as mass shell); the independent-phase spinor completion; V–A odds as cxp/crx after chiral import; the two-angle beta/Coulomb structure with $\xi = \sin a/\sin x$; rational beta phase space; the exact native-carrier characterization. **Strongest negative results retained:** one scalar gains no rank

from many evaluations; chirality orientation not derived; $a \neq \chi$ necessarily; no complex-Gamma from rational closure; direct E/B assignment fails; local scalar duality is pure gauge; NP weights are tetrad-gauge; conformal dressing invisible to source-free Maxwell; helicity sectors not identified by coincidence. Later work must survive these obstructions rather than restart before them.

[← Series index](#) · [Volume 0 overview](#) · [Next: Fermion Composition →](#)

A Volume 0 book of the R Theory rewrite. Source: the former T4 essay-book, rebuilt standalone. Status discipline: exact / conditional / import / obstruction — nothing promoted beyond its support. $\Delta_{\text{op}} = \emptyset$ throughout.

Fermion Composition, Color Closure, and Bound-State Geometry

Volume 0. How fermionic systems compose once one fermion is no longer enough — rebuilt dependency-first, with the failures kept on the record.

About this book. Rebuilt as standalone R Theory from the former T5 essay-book, which was itself a "clean dependency-first reconstruction" of an earlier research draft. Each section states its inputs up front; no later section retroactively supplies an assumption an earlier result needed. The book's two architectures — **color** and **mass/binding** — are proved logically independent of each other.

Status key. **EXACT** proved mathematics · **CONDITIONAL** exact after a declared import or contract · **DECLARED** a mathematical extension, labeled as such · **IMPORT** accepted physics taken as input · **OBSTRUCTION / NEGATIVE** a proved or retained negative result.

Depends on: [Book 0](#) (composition rank, Clifford structure); [Book 20](#) (the half-angle witness); the internal radial Dirac–Coulomb paper (§5).

§1 Firewall

§2 Color

§3 QCD

§4 Two fermions

§5 Bound states

§6 Tests

§7 Failures

§8 E8

§9 Audit

§1 — The composition firewall

Before color, mass, or binding: what does it mean to compose two systems? For real finite-dimensional carriers there is an exact decomposition

$$\text{Sym}(V_A \otimes V_B) = [\text{Sym}(V_A) \otimes \text{Sym}(V_B)] \oplus [\text{Alt}(V_A) \otimes \text{Alt}(V_B)]$$

— the locally symmetric product span is *not* generically the whole composite symmetric sector. A globally symmetric sector ($\text{Alt} \otimes \text{Alt}$) can be invisible to any reconstruction restricted to local symmetric factors. Consequence, kept as a standing obstruction: **a list of locally visible subsystem coordinates cannot be**

assumed to determine the full composite state unless local tomography is separately established. No later success in this book may be used to erase this rank obstruction.

EXACT **Composition firewall.** New axioms: 0. New physics imports: 0. The color weight diagram is not the whole state of a hadron; pair coordinates are not the whole two-fermion state; a constituent list is not a complete interacting record.

§2 — Three states, roots, and $\mathfrak{su}(3)$: the correct logical order

Declare one mathematical extension: **DECLARED** $V_3 = \mathbb{C}^3$ — not derived from Book 0, declared. From it alone: the centered weights w_i form an equilateral triangle in the sum-zero plane; their pairwise differences form the A_2 root system (Cartan matrix $[[2,-1],[-1,2]]$, six roots, regular hexagon); and the commutator chain from one $\mathfrak{su}(2)$ plus one bridge closes to all of $\mathfrak{su}(3)$. One bridge is minimal *within the declared candidate class* — this is not a theorem that nature must realize color this way.

Then the demotion that fixes the logical order: the tetrahedral metric $G = \frac{1}{2}(I+J)$ restricts to a scalar multiple of the Euclidean metric on the A_2 sum-zero plane, so it is an **exact compatible realization** of the A_2 geometry — but A_2 does not uniquely select it and it does not uniquely select A_2 . The correct order is $\mathbb{C}^3 \rightarrow \mathbf{weights} \rightarrow A_2 \rightarrow \mathfrak{su}(3)$, with tetrahedral geometry as an optional realization layer, never a foundation.

Three-state closure **EXACT**

A declared complex three-state carrier has a canonical weight triangle, the A_2 root system, and $\mathfrak{su}(3)$ Lie closure from one bridge. **OBSTRUCTION** Tetrahedral nonuniqueness: the tetrahedral metric is compatible, not foundational. **OBSTRUCTION** $\mathfrak{su}(3)$ -isomorphic mathematics does not derive physical QCD.

§3 — The QCD import: color map, singlets, and what is not derived

Now the physics enters by explicit import. **IMPORT** Standard QCD: quarks in the 3, antiquarks in the $\bar{3}$, gluons in the 8, $\text{su}(3)$ -valued connection — plus a **CONDITIONAL** projection contract identifying the abstract V_3 carrier with the QCD fundamental color carrier. After the import, the weight triangle becomes the red/green/blue Cartan weights ($30^\circ, 150^\circ, 270^\circ$), anticolors are antipodes, and the color/anticolor hexagon interlaces the root hexagon with an exact 30° offset. The standard singlet channels follow exactly: $3 \otimes \bar{3} = 1 \oplus 8$, $3 \otimes 3 \otimes 3 = 1 \oplus 8 \oplus 8 \oplus 10$, with the ε -contraction exactly color-invariant.

Two firewalls stay attached. First, the polar color map lives in the internal (T_3, T_8) Cartan plane — its angles are representation coordinates, **not** spatial directions; no diagram-shape resemblance licenses identifying color axes with x, y, z. Second, and crucially: **singlet representation theory does not derive confinement**. The tensors δ and ε prove invariant channels; they do not prove a linear potential, mass gap, asymptotic freedom, or the absence of colored asymptotic states. Confinement, g_s , Λ_{QCD} , and the Hamiltonian are imported dynamics.

CONDITIONAL **Triadic color closure.** Conditional on the QCD import, three fundamental color carriers possess an exact antisymmetric singlet channel — representation-theoretic closure, not dynamical confinement. **OBSTRUCTION** Neither confinement, nor couplings, nor a Hamiltonian follows from the Lie algebra.

§4 — Two fermions: the exact invariant and its independence

Independently of color — this path uses none of §§1–3 — bring two supplied masses into the half-angle coordinate language. With mass rapidity $\lambda_m = \ln(m_1/m_2)$, $q_m = \tanh(\lambda_m/2)$, relative rapidity η , and $q_v = \tanh(\eta/2)$, ordinary on-shell relativity gives the exact invariant

$$s = 2m_1m_2(\cosh \lambda_m + \cosh \eta)$$

equivalently $s/(m_1m_2) = [R(q_m)+R(q_m)^{-1}] + [R(q_v)+R(q_v)^{-1}]$ with $R(q) = (1+q)/(1-q)$: the same reciprocal transfer architecture appears twice inside one Lorentz scalar — mass asymmetry and relative motion. Two honesty guards are load-bearing: **q_m and q_v are independent physical data** (identifying them would be an extra ansatz, not a consequence), and **writing $m_1/m_2 = R(q_m)$ is a coordinate change, not a mass-generation mechanism** — it explains nothing about why the ratio has its observed value.

Two-fermion invariant EXACT

From supplied masses and ordinary special relativity — no color, no QCD, no tetrahedral geometry. q_m is not a Lorentz rapidity; repeated Cayley form does not reduce physical rank; no force, potential, or binding law appears; no particle mass is predicted.

§5 — Below threshold: the bound-mass coordinate and its conditional convergence

Continue the invariant below the free two-particle threshold on a declared analytic branch DECLARED

$\eta \rightarrow i\theta$, and set $u = \tan(\theta/2)$. Solving gives the exact bound-mass coordinate

$$u^2 = [(m_1+m_2)^2 - M^2] / [M^2 - (m_1-m_2)^2], \quad M^2 = (m_1-m_2)^2 + 4m_1m_2/(1+u^2)$$

— an invertible reparameterization of a supplied subthreshold mass, from which no force, spectrum, or binding dynamics follows. Weak binding gives $u^2 = B/(2\mu) + O(B^2)$, so the reduced mass appears as the leading conversion scale with no separate ansatz. The heavy-source limit gives $u^2 \rightarrow (m_2-E)/(m_2+E)$.

Then the book's strongest structural result. Import the standard point-Coulomb Dirac equation and the radial Dirac–Coulomb paper's circular-state theorem: on the $n_r = 0, \kappa < 0$ heavy-source branch, the radial spinor ratio is constant and $|G/F|^2 = (m-E)/(m+E)$ — the identical ratio. Hence, with no particle-mass ratio fitted,

$$u_{\text{pair}} = |G/F| = \text{sxp}(x)$$

exactly, on a declared state family, from two independent routes. The failure domain is kept visible: for $n_r > 0$ the ratio runs with radius (Prüfer phase plus independent amplitude), so **one global u cannot reconstruct a generic excited radial spinor.**

Independent coordinate convergence CONDITIONAL

Exact on the circular Dirac–Coulomb branch, conditional on imported Coulomb and Dirac–Coulomb physics. Not an empirical deviation from that physics. The $n_r > 0$ obstruction is part of the theorem's

statement, not a footnote.

§6 — Observational discipline: thresholds, transitions, and verified numbers

What actually determines u ? An absolute breakup threshold fixes it uniquely; a single transition frequency constrains only a relation between two absolute coordinates — and **increasing precision does not remove that rank deficiency**. The preferred control ladder is positronium $\rightarrow$ muonium $\rightarrow$ hydrogen $\rightarrow$ deuteron $\rightarrow$ heavier systems only with declared constituent masses. The numbers check (recomputed 2026-09-19 from the frozen inputs):

```
Hydrogen:  q_m ≈ 0.99891135885,  u_H(1S) ≈ 0.003648720323
            α/2 ≈ 0.003648676282 < u_H < u_C ≈ 0.003648724858
            (u_C = infinite-source circular Dirac coordinate)
Deuteron:   B_d ≈ 2.22456637 MeV,  q_m ≈ 0.00068873505,
            u_d ≈ 0.0487186123  (≈13× the atomic Coulomb scale)
Muonium:    q_m ≈ 0.99037 (leptonic, far from positronium's 0)
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EXACT **Deuteron control.** The deuteron sits near $q_m \approx 0$ while its u exceeds the atomic Coulomb scale by more than an order of magnitude — q_m and u are empirically independent, and no universal $u = \alpha/2$ law survives. **OBSTRUCTION** A difference from $\alpha/2$ or the infinite-source limit is not a Projection residual until the reference theory is matched at the same precision. **OBSTRUCTION** Scheme-undefined light-quark constituent masses do not define an invariant u .

Preregistration rule. Before testing any new mass law, record: constituent definition, bound-state definition, coordinate extraction, named reference theory, independent Projection prediction, parameter provenance, uncertainty propagation, invariant observable difference Δ_{op} , and failure criterion. Without these, the exercise is coordinate analysis, not empirical promotion. A residual may be read as a closure defect only when two independently specified theoretical routes produce the same observable — **measurement is not a second theory path**.

§7 — The frozen failure record

The original research program behind this book tried to find mass laws and failed. The failures are retained here as part of the book, not tuned away:

- **81/32 is not exact.** $R_\beta = (m_n - m_p)/m_e \approx 2.53098858276$ versus $81/32 = 2.53125$ — a residual of about -133.6 eV, many standard deviations from exact equality. It must not be presented as an exact mass formula.
- **Fifth-lattice proximity is not mass quantization.** Near-hits exist; negative lepton controls reject a universal all-masses-on-the-lattice rule, and same-isospin $d \rightarrow u$ tests reject the stronger claim.
- **Closure defects are not musical commas.** Coleman–Glashow and Gell-Mann–Okubo defects do not reproduce a clean Pythagorean or syntonic-comma law.
- **The order-five mass route failed.** An exact algebraic grading of chiral sectors does not generate observed fermion masses; restoring a Yukawa term needs imported scalar structure, and the grading then supplies no independent prediction. No new low-energy selection rule was promoted.

These negatives are inherited by any future flavor/mass program. They may not be silently reversed.

§8 — Optional appendix: the higher-carrier audit (firewalled)

The E8(−24) higher-carrier work is preserved here under its own dependency firewall: **no theorem in §§1–7 depends on this appendix; deleting it changes nothing above.** The honest negatives are kept: exact matrix particle/mirror exchange is not a physical oscillation law without dynamics; equal cyclic order is not identity of automorphism action; no fermion mass spectrum has been derived. The positive structural claims — the 248-direction reconstruction, the zero-residual particle/mirror matching, the $su(2,3) \oplus u(1)_Y$ Standard Model centralizer, the commuting SU(5) partner — are cited to the project's Verification Ledger, **not proved in this text**, and cannot be stated as established until that ledger and its code are rerun. They are recorded here as ledger-cited, nothing more.

§9 — Closure audit

Book 21 promotion audit

Projection-specific new equation of motion: none.

Independently determined coupling or mass parameter: none.

New invariant observable beyond invertible rewrites: none.

Quantitative deviation from a named reference theory: none predicted.

Therefore $\Delta_{\text{op}}(\text{Book 21}) = \emptyset$. Null hypothesis C0 retained.

Book 21 closure EXACT

Relative to the inherited foundations and the declared imports, this book establishes: the composition firewall; three-state $\rightarrow A_2 \rightarrow \text{su}(3)$ closure with the tetrahedral metric demoted to realization; conditional triadic color closure with confinement kept as imported dynamics; the exact two-fermion invariant with q_m, q_v independent; the below-threshold coordinate with its reduced-mass and heavy-source limits; the conditional Dirac–Coulomb convergence $u = |G/F| = \text{sxp}(x)$ on the circular branch; verified hydrogen/deuteron/muonium controls; the frozen failure record; and the firewalled higher-carrier audit. The color and mass architectures are logically independent — they do not derive one another. $\Delta_{\text{op}} = \emptyset$; C0 survives.

Dropped in this rebuild. The "closure reset" vocabulary (honestly labeled in the source but adding no dynamics — its own negative theorem says so) and the baryon–lepton "research lead" (an unpromoted hypothesis) are not carried into the main text. Open leads live outside the promoted ledger.

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A Volume 0 book of the R Theory rewrite. Source: the former T5 essay-book (v2 reconstruction), rebuilt standalone. Status discipline: exact / conditional / declared / import / obstruction — nothing promoted beyond its support. $\Delta_{\text{op}} = \emptyset$ throughout.

Canonical Spin–Geometry and the Primitive Symplectic Atlas

Volume 0. The volume's strongest exact result: the four primitives are canonical charts on one radial-spinor phase plane — and the audit that keeps the imported machinery honest.

About this book. Rebuilt as standalone R Theory from the former T6 essay-book, the "canonical" book of the Projection Craft series — its declared job was to make the primitive functions canonical coordinates on the radial-spinor phase space. That core is proved. Around it sits heavy imported machinery (Dirac–Coulomb theory, the first-order Einstein–Hilbert action, Maxwell, rotation, Einstein–Cartan, Kerr), and this book labels every layer: what is proved internally, what is conditional on import, and what is not established here. The "promotion" of earlier results is explained precisely: it means *restriction to a specialized setting and reuse of existing proofs* — never a status upgrade and never new evidence.

Status key. **EXACT** proved mathematics · **CONDITIONAL** exact after a declared import or contract · **IMPORT** accepted physics taken as input, not derived · **OBSTRUCTION** a proved or retained negative result.

Depends on: [Book 0](#); [Book 20](#) (§4's two-angle structure); the internal radial Dirac–Coulomb paper.

§1 Spin transport

§2 Prüfer pair

§3 Symplectic atlas

§4 Defect action

§5 Gauge & rotation

§6 Kerr

§7 Spin obstruction

§8 Audit

§1 — Radial coframe and spin transport (imported machinery, labeled)

The book's canonical apparatus starts from accepted background: **IMPORT** the radial coframe e^a (flat at $\eta=0$, with defect $\delta_e = e^0 - dr$), the torsion-free spin connection, and the Dirac spinor Ψ in the projected chart. Two declared contracts carry the whole book: **C6a**, the radial two-component chart $\Psi \rightarrow (r, F, G, \dots)$, and **C6b**, that F and G are independent on open sets. Two obstructions are kept permanently: the

spherical-to-Cartesian map is not a global diffeomorphism (origin and polar axis excluded), and C6a/C6b fix the *reading* of the geometry but *not* the physical meaning of F and G — importing the radial Dirac–Coulomb paper promotes the chart, never the book's own assumptions.

§2 — The Prüfer canonical pair (exact conditional)

The transport operator on (F, G) has exact zero trace, which yields the exact radial invariant

$$F F' + G G' = -A^2/(2r)$$

and a canonical pair: the Prüfer angle $\theta = \arg(G + iF)$ and the spin density $A^2 = F^2 + G^2$. Under C6a/C6b the pair is regular on open sets and the spin phase space carries

$$\omega_{\text{spin}} = 4 \, dF \wedge dG$$

— exactly canonical. This is exact *after* importing the radial Dirac dynamics; the symplectic form is a conditional theorem, not a derivation of Dirac theory.

Canonical spin phase space CONDITIONAL

Under C6a/C6b and the imported radial Dirac equations, (Θ, A^2) is a canonical pair on the radial-spinor phase plane with symplectic form $4 \, dF \wedge dG$. OBSTRUCTION Does not derive the Dirac equation, spin- $\frac{1}{2}$, the spin-statistics connection, or the fine-structure constant.

§3 — The primitive symplectic atlas (the volume's strongest exact result)

The centerpiece. The four primitive functions are **canonical charts on the same (F, G) plane** — four Darboux-compatible coordinate systems with a common spin one-form:

$$p_{\text{sxp}} = 2F^2, \quad p_{\text{srx}} = -2G^2, \quad p_{\text{cxp}} = (F-G)^2, \quad p_{\text{crx}} = -(F+G)^2$$

$$p_Q dQ = -2G dF + 2F dG$$

and the common symplectic form $\omega_{\text{spin}} = 4 dF \wedge dG$, already established in §2. The atlas maps cover all four quadrants; the reciprocal and complementary identities are exact differential identities. On the circular Dirac–Coulomb branch the atlas charts meet the radial paper's spinor-ratio theorem, so the atlas variables and the spinor-ratio results are cross-checked against the same imported dynamics. No new physics is claimed: the primitives were already canonical coordinate expressions; the atlas shows they are canonical *together*.

Primitive symplectic atlas **EXACT**

The four primitive momentum charts with common one-form and common symplectic form constitute an exact internal theorem, conditional only on the declared C6a/C6b chart and the imported radial dynamics. The chart transition identities, reciprocal relations, and amplitude-ratio readings are exact. This is a representation result: it changes no prediction of the imported theory.

§4 — The first-order spherical defect action (conditional reconstruction)

Now the imported gravitational machinery. **IMPORT** the first-order Einstein–Hilbert action in the spherical defect variables $(\delta_e, \delta_f, \delta_\omega)$, with a regularized radial measure $\eta^2(r)dr$ on (r_{UV}, ∞) and the standard asymptotically flat falloff conditions declared. Exact reductions under these imports: the gravitational defect kinetic structure; the boundary term; and the vanishing of the on-shell defect action in the massless flat defect-free limit. What remains outside the reduction: the regularization contract, the falloff conditions, and the accepted Einstein–Hilbert machinery itself — conditional reconstruction, not a derivation of GR. The mass parameter m appearing in the defect action is imported through the contract; the defect action does not compute it.

CONDITIONAL **Spherical defect reduction.** Exact reduction of the imported first-order Einstein–Hilbert action in the defect variables under declared regularization and falloff. **OBSTRUCTION** Does not derive the Einstein field equations, Newton’s constant, or the mass parameter.

§5 — Gauge, rotation, and spin: conditional channels

Three more channels, each exact after its own import:

- **Gauge (Maxwell as a channel):** after the gauge import, the radial electric charge appears as the cyclic radial momentum p_Φ conjugate to the radial gauge phase — the Maxwell field-strength dynamics and the coupling e are imported; the book supplies the canonical reading, not the theory.
- **Rotation:** angular momentum enters as the canonical momentum p_Ω conjugate to frame dragging — exact in the imported Kerr-geometry channel.
- **Spin (Einstein–Cartan):** the axial contorsion channel is exactly auxiliary: the torsion equation is algebraic and the torsion is solved out in favor of the spin current — so the spin sector carries no independent propagating degree beyond the imported one.

Each channel is a representation of imported physics in the book's canonical coordinates. None derives its import.

§6 — Kerr/Kerr–Newman half-angle identities (with guards)

On the imported Kerr geometry, the horizon radii satisfy exact half-angle identities in the book's coordinates — and the same identities extend to Kerr–Newman after the electromagnetic import. The guards are part of the result: the identities live on the declared equatorial/chart domain (coordinate singularities excluded), they are consequences of the imported metric, and they establish no new black-hole physics. The spherical-spin obstruction (§7) is the reason these identities do not become a spin law.

CONDITIONAL **Kerr/Kerr–Newman identities.** Exact coordinate identities on the imported geometries, valid on the declared regular domain. **OBSTRUCTION** No new black-hole physics; no spin dynamics derived.

§7 — The spherical-spin obstruction

The book's hardest negative, and its most load-bearing. The spherical symmetry used throughout forces the total spin current to integrate to zero over the sphere — a spherically symmetric configuration cannot carry net spin. The canonical spin apparatus of §§1–3 therefore **cannot be promoted to a theory of spin in the spherical sector**. This is not a defect of the atlas; it is a theorem about the domain: the spin variables are exact internal coordinates of the radial-spinor phase plane, and the spherical obstruction says exactly where they stop being physics. Any future spin program must begin by leaving spherical symmetry.

OBSTRUCTION **Spherical spin cancellation.** Spherical symmetry kills net spin current. The atlas is exact where it applies and silent where it cannot speak.

§8 — The does-not-establish ledger and book closure

The source book's §6.XV is preserved verbatim in spirit: the does-not-establish ledger. This book does not derive: the Dirac equation; spin- $\frac{1}{2}$; the fine-structure constant; the Coulomb interaction; the Einstein field equations; Newton's constant; the Maxwell equations; the charge e ; the Kerr metric; the Einstein–Cartan spin coupling; particle masses; couplings; or new observables. "Promotion" of earlier results (Books 20–21, the radial paper) means restriction to a specialized setting with reuse of existing proofs — a status-preserving operation, never a status upgrade and never new evidence. The Yang–Mills attribution from the source preamble is **not** repeated: the cited result is absent from the cited text (see the [Volume 0 provenance note](#)).

Book 22 promotion audit

Exact internal theorems:	spin invariant, Prüfer pair, primitive symplectic atlas, atlas transition identities.
Exact after declared import:	Kerr/KN identities, defect-action reduction, gauge/rotation/spin channels.
New equation of motion:	none.
New coupling or mass parameter:	none.
New invariant observable:	none.

Prediction difference: $\Delta_{\text{op}} = \emptyset$. C0 retained.

Book 22 closure EXACT

The four primitives are canonical charts on one radial-spinor phase plane — the primitive symplectic atlas is the strongest exact result of Volume 0, and it changes nothing it did not already change: it is a representation theorem inside imported physics. With the obstructions kept (spherical spin cancellation, coordinate-domain guards, the does-not-establish ledger), $\Delta_{\text{op}} = \emptyset$ and C0 — *Projection Craft changes representation but not physics* — is retained, undefeated, and still falsifiable.

Forward boundary. The book's own reopening rule: C0 is defeated only by a future fully specified Class-C model with a stated measurement contract — a new invariant observable that survives the redundancies this volume catalogued. The spherical-spin obstruction marks where any spin program must start: outside spherical symmetry.

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